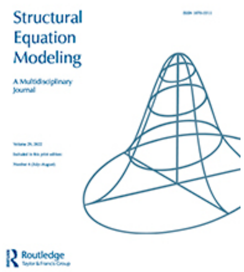

Structural Equation Modeling: A Multidisciplinary Journal

ISSN: 1070-5511 (Print) 1532-8007 (Online) Journal homepage: www.tandfonline.com/journals/hsem20

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To cite this article: Gyeongcheol Cho, Heungsun Hwang, Marko Sarstedt & Christian M. Ringle (2022) A Prediction-Oriented Specification Search Algorithm for Generalized Structured Component Analysis, Structural Equation Modeling: A Multidisciplinary Journal, 29:4, 611-619, DOI: 10.1080/10705511.2022.2057315

To link to this article: https://doi.org/10.1080/10705511.2022.2057315

Published online: 19 Apr 2022.

STRUCTURAL EQUATION MODELING: A MULTIDISCIPLINARY JOURNAL
2022, VOL. 29, NO. 4, 611–619
https://doi.org/10.1080/10705511.2022.2057315

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A Prediction-Oriented Specification Search Algorithm for Generalized Structured Component Analysis

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ABSTRACT

Generalized structured component analysis (GSCA) is used for specifying and testing the relationships between observed variables and components. GSCA can perform model selection by comparing theoretically established models. In practice, however, theories may not always completely and unambiguously specify the relationships between variables in the model. In such situations, a specification search strategy allows for exploring potential relationships between variables in a data-driven manner. A specification search based on prediction of unseen observations is attractive as it does not require the provision of theoretically plausible models. To date, GSCA has not been equipped with such a specification search strategy. Addressing this limitation, we propose a prediction-oriented specification search algorithm for GSCA, which reveals the best combination of predictors that minimizes each target variable's prediction error. We conduct a simulation study to examine the new algorithm's performance and apply it to real data to further investigate and demonstrate its practical usefulness.

KEYWORDS

Cross-validation; generalized structured component analysis; predictive modeling; specification search

1. Introduction

Generalized structured component analysis (GSCA; Hwang & Takane, 2004, 2014) is a general multivariate method for specifying and testing the relationships between observed variables and components in a theoretically plausible manner. A component is a linear combination or weighted sum of observed variables that may serve as an index that summarizes and ranks specific observations (Cho et al., 2021; Rigdon, 2022). For instance, socioeconomic status (SES) is typically represented by a weighted sum of education, income, and occupation (American Psychological Association, 2007). Likewise, in genomic research, a gene is considered a biological cluster of numerous single nucleotide polymorphisms (SNPs) that occur within the gene, so that researchers often represent a gene as a weighted sum of its SNPs (e.g., Dai et al., 2013; Horne & Camp, 2004; Hwang et al., 2021).

Given that observed variables' scores are linearly independent (Lay et al., 2015, Chapter 1, Theorem 11), GSCA can provide unique, deterministic component scores for each individual observation, whose computation draws on the observed variables' weights obtained as part of parameter estimation. In addition, the component scores for new observations can be computed by post-multiplying their scores of the observed variables by the weight estimates. These component scores can be utilized for predicting unseen values of target variables of interest, which can be observed variables or components in the model, or evaluating the model's prediction accuracy for the target variables. Accordingly, GSCA is well-suited for predictive modeling that aims at building a model that can best predict new observations on target variables, using individual component scores (Cho et al., 2019, 2022; Hwang, et al., 2020).

GSCA is currently able to perform prediction-oriented model selection. This procedure requires a set of theoretically established models in advance. A cross-validation index named the out-of-bag prediction error (OPE; Cho et al., 2019) determines the predictive power of each of these models. The OPE is calculated for each model by cross-validating sets of training and validation samples that are constructed from the original sample via bootstrapping (Efron, 1979). The theoretically established model with the smallest OPE value is considered the best predictive model in the set.

In practice, however, theories may not always be available to completely specify the relationships between variables (observed variables or components), especially when researchers are at an early stage of theory development. When theoretical support and domain knowledge are scarce, researchers may conduct a specification search (Long, 1983; MacCallum, 1986) which explores the structural relationships between variables in a statistical or data-driven fashion. Particularly, a prediction-oriented specification search strategy, which is based solely on prediction of unseen observations, can be attractive for researchers as it does not require the provision of the correct model to ensure the model search's legitimacy (Shmueli, 2010; Shmueli & Koppius, 2011). A specification search of this type focuses on testing the theoretical falsifiability of different models

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(Popper, 1962) — repeatedly over test samples that are not used for estimating the parameters of the model. Despite the obvious usefulness of this approach, no prediction-oriented specification search strategy has yet been developed for GSCA.

To address this issue, we propose a prediction-oriented specification search algorithm for GSCA, named *predictive feedforward* (PF). In the PF algorithm, researchers do not need to completely specify the relationships between variables in advance. Instead, they can decide which variables are target variables and which variables are candidate predictors for the target variables. Then, the algorithm automatically searches for the best combination of predictors, which minimizes the target variables' expected prediction error. The result represents a key orientation point for researchers in the development of their final theoretically established model.

In the sections that follow, we provide a brief overview of GSCA and describe the proposed model search algorithm in detail. Then, we conduct a simulation study to investigate the performance of the algorithm and report the results. Finally, we apply the algorithm to real data to further investigate and demonstrate its practical usefulness before concluding with the main findings and contributions of this research.

2. Method

2.1. Overview of GSCA

To facilitate an understanding of the proposed algorithm, we begin by briefly explaining model specification and parameter estimation in GSCA. Let $\mathbf{\Gamma} = [\gamma_1, \ldots, \gamma_P]$ denote an N by P matrix of components, where γ_p is an N by 1 vector of the pth component ($p = 1, \ldots, P$) and N is the number of observations. Let $\mathbf{Z} = [\mathbf{z}_1, \ldots, \mathbf{z}_J]$ denote an N by J matrix of observed variables or indicators, where $\mathbf{z}_j$ is an N by 1 vector of the jth indicator ($j = 1, \ldots, J$). Let $\mathbf{C}$ denote a P by J matrix consisting of loadings that relate indicators to components. Let $\mathbf{B}$ denote a P by P matrix of path coefficients relating components among themselves. Let $\mathbf{W}$ denote a J by P matrix of component weights assigned to indicators. Let $\mathbf{E}_Z$ denote an N by J matrix of errors for indicators. Let $\mathbf{E}_\Gamma$ denote an N by P matrix of errors for components. Both indicators and components are assumed to be standardized (i.e., $\gamma_{p'}\gamma_p = \mathbf{z}_{j'}\mathbf{z}_j = N$).

In model specification, GSCA involves three-sub models, measurement, structural and weighted relation models, as follows (Hwang & Takane, 2014, Chapter 2).

$$\mathbf{Z} = \mathbf{\Gamma C} + \mathbf{E}_Z \tag{1}$$

$$\mathbf{\Gamma} = \mathbf{\Gamma B} + \mathbf{E}_\Gamma \tag{2}$$

$$\mathbf{\Gamma} = \mathbf{ZW} \tag{3}$$

The measurement model (1) is used to specify the relationships between indicators and components, and the structural model (2) is to express the relationships among components. The weighted relation model (3) defines each component as a weighted composite of indicators, explicitly indicating that GSCA is a component-based method.

The three sub-models are integrated into a single equation, called the GSCA model, as follows:

$$\begin{aligned}[\mathbf{Z}, \mathbf{ZW}] &= \mathbf{ZW}[\mathbf{C}, \mathbf{B}] + [\mathbf{E}_Z, \mathbf{E}_\Gamma] \\ \mathbf{ZV} &= \mathbf{\Gamma A} + \mathbf{E},\end{aligned} \tag{4}$$

where $\mathbf{V} = [\mathbf{I}, \mathbf{W}]$, $\mathbf{A} = [\mathbf{C}, \mathbf{B}]$, and $\mathbf{E} = [\mathbf{E}_Z, \mathbf{E}_\Gamma]$. The model parameters in $\mathbf{W}$, $\mathbf{C}$, and $\mathbf{B}$ are estimated by minimizing the following least squares objective function.

$$\phi = \mathrm{SS}(\mathbf{E}), \tag{5}$$

where SS() stands for the sum of squares. An alternating least squares algorithm is typically used to minimize this objective function (Hwang & Takane, 2004, 2014). GSCA estimates the parameters without distribution assumptions such as multivariate normality of indicators. It obtains the standard errors or confidence intervals of the parameter estimates nonparametrically via the bootstrap method.

2.2. Predictive Feedforward Specification Search Algorithm for GSCA

The proposed algorithm aims to search for the relationships between variables, which show the highest prediction power for target variables, when researchers lack sufficient knowledge for fully specifying the relationships. Prior to applying the algorithm, researchers need to decide which variables serve as target and which are used to predict each target variable. For selected target variables, the proposed algorithm proceeds with the following three steps.

Step 1: The algorithm begins by fitting a default model with which researchers wish to start. This model, denoted by M_1, can be the null model without any predictors or a model with some predictors. Note that M_1 shall generally represent the baseline model to be compared in terms of prediction error during the execution of the algorithm. Then, the algorithm estimates M_1's expected prediction error via K-fold cross-validation (Hastie et al., 2001, pp. 241–249). In K-fold cross-validation, a sample is divided into K sub-samples. One of the subsamples becomes the validation sample whereas the remaining ones are used as the training sample. A model is fitted to the training sample and its parameters are estimated. Then, with the parameter estimates fixed, the algorithm computes a prediction error of the model based on the validation sample. This procedure is repeated over K pairs of the training and validation samples. Next, the cross-validation estimate of expected prediction error for target variables (CVPE_T) is calculated as follows:

$$\mathrm{CVPE}_\mathrm{T} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{QN_k}\mathrm{SS}\left([\mathbf{Z}_k\hat{\mathbf{V}}_k - \mathbf{Z}_k\hat{\mathbf{W}}_k\hat{\mathbf{A}}_k]\mathbf{O}\right), \tag{6}$$

where Q is the number of target variables, N_k is the number of observations in the kth validation sample ($k = 1, \ldots, K$), $\mathbf{Z}_k$ is an N_k by J matrix of indicators in the kth validation sample, $\hat{\mathbf{W}}_k$ consists of the weights estimated from the kth training sample, $\hat{\mathbf{A}}_k$ includes the loadings and path coefficient estimates from the kth training sample, and $\mathbf{O}$ is a Q by Q diagonal matrix whose qth diagonal entry is one if the qth variable is chosen as a target variable and zero otherwise ($q = 1, \ldots, Q$). In (6), $\mathbf{Z}_k$ should be standardized based on the sample means and variances of indicators in the kth training sample.

Step 2: The algorithm considers a set of competing models, each of which adds one distinct predictor for the target variables to M_1. It fits all the competing models, computes their $CVPE_T$ values, and selects the model with the smallest $CVPE_T$, which is then denoted by M_2. If the $CVPE_T$ value of M_2 is smaller than that of M_1, the algorithm tests if the $CVPE_T$ difference between M_1 and M_2 is statistically significant based on the nonparametric bootstrap method. Specifically, it produces the 95% nonparametric bootstrap confidence interval of the $CVPE_T$ difference over H bootstrap samples. If this confidence interval does not include a zero, the $CVPE_T$ difference between M_1 and M_2 is statistically significant. If the $CVPE_T$ difference is statistically significant, M_2 is now chosen as the new baseline model M_1, replacing the previous one. Subsequently, Step 2 is applied to another set of competing models, each of which includes one of the remaining predictors that are not used in M_1. This process is repeated for the target variables until the $CVPE_T$ of M_2 is greater than that of M_1, the $CVPE_T$ difference between M_1 and M_2 is statistically nonsignificant, or M_1 has considered all candidate predictors for the target variables.

Step 3: The M_1 being left upon the algorithm's completion is selected as the final model with the best prediction power for the target variables.

Beyond these explications, some comments on the PF algorithm are warranted. First, the PF algorithm applies the nonparametric bootstrap method to the original sample only once and continues to use the same bootstrap samples for comparing the $CVPE_T$ difference between two models. This avoids the possibility that their $CVPE_T$ difference stems from the use of different sets of bootstrap samples. Second, although the algorithm appears to proceed in a manner similar to forward-selection stepwise regression, it does not compare two models in terms of their explanation power for the sample at hand (e.g., R^2 or F statistic), which has been criticized for potential over-fitting and reliance on p values that can be reliable only when the fitted model is true (Harrel, 2001, p. 4; Smith, 2018). Instead, the algorithm compares two models in terms of their expected prediction error of target variables' unseen values, which is estimated by the $CVPE_T$. Third, researchers need to decide on K before applying the algorithm. When K is large, $CVPE_T$ has a smaller bias in the true prediction error but may have a higher variance. In contrast, a small K may have a smaller variance but a larger bias. In practice, $K = 5$ or 10 is typically recommended (Breiman & Spector, 1992; Hastie et al., 2001, pp. 241–243; Kohavi, 1995). As the number of bootstrap samples (H) becomes larger, the bootstrap confidence interval of the $CVPE_T$ difference will be more accurately estimated. However, researchers should also consider computational burden in the selection of H. As shown in our subsequent simulation study, we find that $H = 100$ seems to perform well to identify a model with little false predictors. Fourth, the algorithm allows researchers to choose any indicator or component as a target variable in the model and calculate the model's $CVPE_T$ as shown in (6). If all indicators and components are chosen as target variables and $K = N$, then $CVPE_T$ becomes equivalent to OPE (Cho et al., 2019; Hastie et al., 2001, pp. 592–593). Lastly, all the steps of the algorithm do not require any distribution assumption of indicators (e.g., normality). This is consistent with GSCA that does not rely on a distributional assumption of indicators for estimating parameters and testing their statistical significance.

3. Simulation Study

We conduct a simulation study to investigate the performance of the proposed model search algorithm. Our study considers a complex population model with two dependent and ten independent components, each of which is associated with three indicators (Figure 1). For each component, we determine component loadings as .6, .7, and .8, and component weights as .4027, .4698, and .5396, as in previous studies (e.g., Cho et al., 2021). We consider the following path coefficients among γ_1, γ_2, γ_3, γ_{11}, and γ_{12} in the structural model: $b_1 = .2$, $b_2 = .4$, $b_3 = .6$, $b_4 = .2$, and $b_5 = .6$. These path coefficients are to reflect small, medium, or large effects. All other path coefficients are set to zero. Finally, we set all the correlations among exogenous components (i.e., $\gamma_1 - \gamma_{10}$) to be .3.

We consider five levels of sample size ($N = 50$, 100, 250, 500, and 1,000), for each of which we generated 500 datasets from the model-implied correlation matrix derived based on Cho and Choi (2020) procedure. We set $K = 5$ and $H = 100$ and choose two components (γ_{11} and γ_{12}) as target variables and use γ_1 to γ_{10} as candidate predictors for γ_{11} and γ_1 to γ_{11} as candidate predictors for γ_{12}. In our default model, all the weights and loadings are correctly specified, whereas none of the path coefficients is specified. We assess the algorithm efficacy in terms of its ability to identify the true structural model (i.e., distinguish between zero and non-zero path coefficients). We apply the PF algorithm to each simulated dataset per sample size. As a benchmark for evaluating the performance of the proposed algorithm, we also apply the tabu search algorithm, which has been adopted for a specification search in structural equation modeling (G. A. Marcoulides et al., 1998). For the tabu search algorithm, we consider the tabu size of 5 and fix the maximum number of iterations to 30, as in Marcoulides and Falk (2018), while using $CVPE_T$ as a model evaluation metric.

Panel A in Figure 2 displays the two algorithms' average rates of selecting the true non-zero path coefficients — the average true positive rates (i.e., how many non-zero path coefficients are on average selected among a total of five non-zero path coefficients), along with the average rates' 95% confidence intervals, over 500 datasets per sample size. The results show that when the sample size is very small ($N = 50$), the PF algorithm can detect about 50% of the

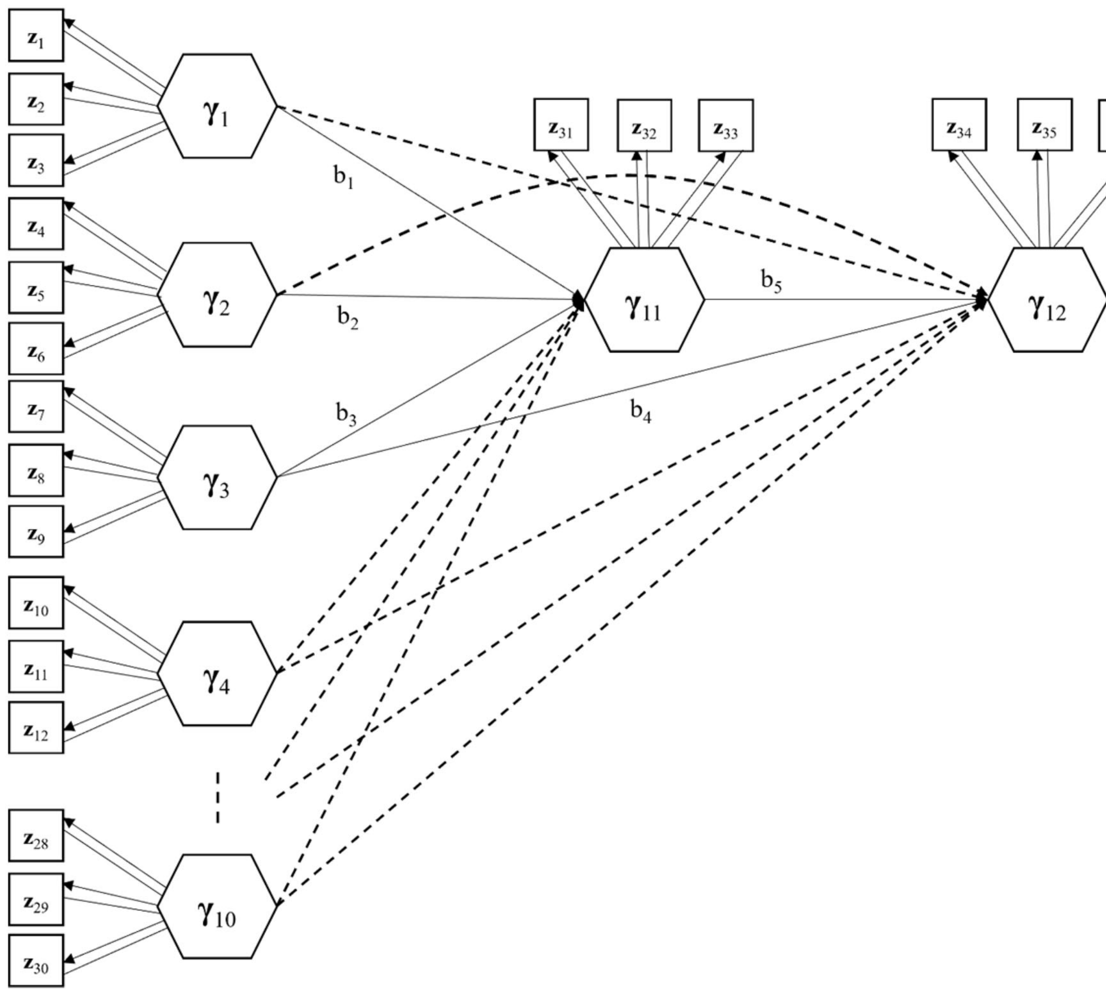

Figure 1. The specified model for the simulation study. *Note:* Hexagons and squares represent components and indicators, respectively. Straight lines indicate weights, arrows denote loadings or non-zero path coefficients, and dashed arrows indicate zero path coefficients. Error terms are omitted.

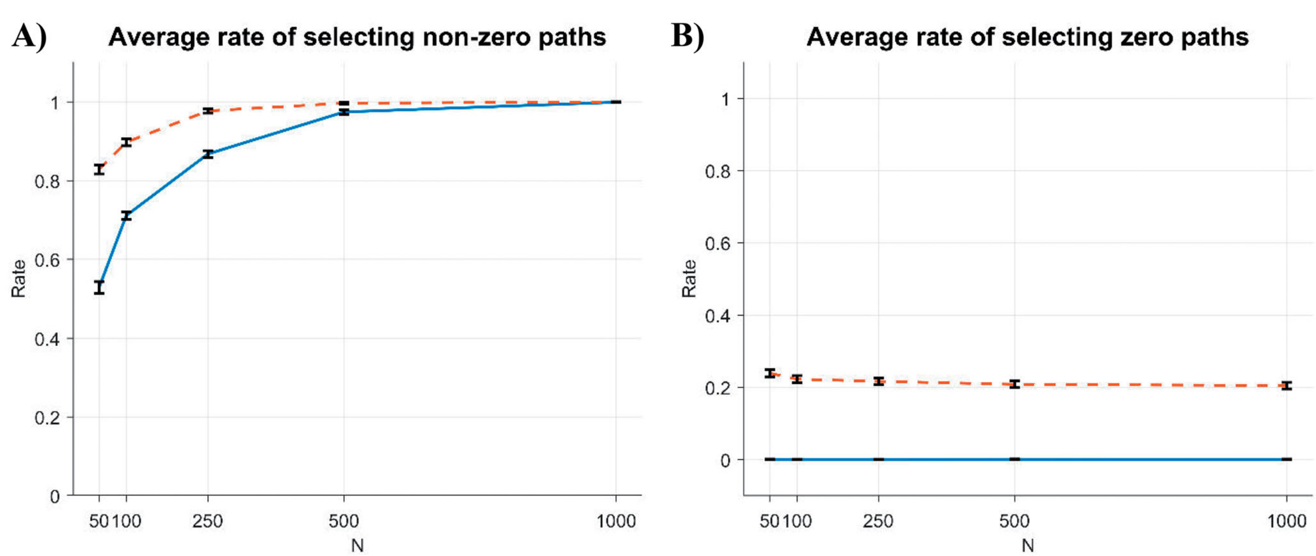

Figure 2. Two search algorithms' average rates of selecting non-zero (true) and zero (false) path coefficients and the average rates' 95% confidence intervals per sample size in the simulation study (Proposed algorithm: —, Tabu algorithm: - - -).

non-zero path coefficients (i.e., around 2.5 non-zero path coefficients) on average. This average true positive rate is lower than that of the tabu search algorithm (i.e., 80%). As the sample size increases, the average true positive rates of both algorithms increase. When $N = 1{,}000$, overall, both algorithms can detect all non-zero path coefficients. Table 1 displays the algorithms' average true positive rates of selecting each individual non-zero path coefficient.

Table 1. Average rate of selecting each non-zero path coefficient per sample size in the simulation study.

		$b_1 = .2$	$b_2 = .4$	$b_3 = .6$	$b_4 = .2$	$b_5 = .6$
Proposed algorithm	$N = 50$	0.11	0.70	0.91	0.12	0.79
	$N = 100$	0.53	0.99	1.00	0.06	0.98
	$N = 250$	0.99	1.00	1.00	0.35	1.00
	$N = 500$	1.00	1.00	1.00	0.87	1.00
	$N = 1{,}000$	1.00	1.00	1.00	1.00	1.00
Tabu search algorithm	$N = 50$	0.82	0.99	1.00	0.44	0.90
	$N = 100$	0.96	1.00	1.00	0.54	0.98
	$N = 250$	1.00	1.00	1.00	0.89	1.00
	$N = 500$	1.00	1.00	1.00	0.98	1.00
	$N = 1{,}000$	1.00	1.00	1.00	1.00	1.00

Panel B in Figure 2 exhibits the algorithms' average rates of selecting false zero path coefficients — the average false positive rates (i.e., how many zero path coefficients are included in the final models among a total of 16 zero path on average) and their 95% confidence intervals per sample size. The average false positive rate of the PF algorithm is virtually zero across all the sample sizes, indicating that the algorithm can exclude all the zero path coefficients in all the conditions. In contrast, the average false positive rate of the tabu search algorithm is close to 20% across all the sample sizes, indicating that the tabu search algorithm tends to favor models that additionally include three or four zero path coefficients on average.

In summary, the simulation study shows that the final models selected via the PF algorithm tend to be parsimonious in that they contain the majority of non-zero path coefficients when sample sizes are small. At the same time, the selected models of the proposed algorithm converge to the specified model with all non-zero path coefficients as sample size increases. Importantly, the PF algorithm does not select any model with false non-zero path coefficients. Taken together, the PF algorithm is able to identify the specified model well. This also suggests that the use of $H = 100$ appears to be acceptably efficient. On the other hand, the tabu search algorithm tends to detect non-zero path coefficients better than the PF algorithm when the sample size is not large (e.g., $N < 500$), although these true positive rates of both algorithms become convergent with the sample size. However, the tabu search algorithm always seems to favor models that contain several false zero path coefficients, resulting in high false positive rates across all the sample sizes. These consistently high false positive rates of the Tabu search algorithm are likely related to its high true positive rates.

4. Empirical Illustration

4.1. UTAUT Model

The present example stems from an empirical study of validating the unified theory of acceptance and use of technology (UTAUT) model (Al-Gahtani et al., 2007) in the context of non-Western cultures. Figure 3 displays the UTAUT model, in which two components, such as users' behavior intention (BI) and their subsequent use behavior (USE) for new IT technology, are assumed to be affected by four components—performance expectancy (PE), effort expectancy (EE), social influence (SI), and facilitating condition (FC). Venkatesh et al. (2003) present a more detailed model description. The dataset used, which we obtained from the SmartPLS website (https://www.smartpls.com/documentation/sample-projects/utaut), has a sample size of 772.

We apply the PF algorithm to identify the best predictive relationships among the components. For comparison, we also apply the tabu search algorithm. Figure 4 displays the relationships among the components that we consider for the specification search algorithms. We treat BI and USE as target components and use {PE, EE, SI, FC} as a set of candidate predictors for BI and {PE, EE, SI, FC, BI} as a set of candidate predictors for USE. As in the simulation study, we apply $K = 5$ and $H = 100$ for the PF algorithm and use 5 as the tabu size and 30 as the maximum number of iterations for the tabu search algorithm.

The two search algorithms select the same final model. The selected model includes all the original path coefficients of the UTAUT model (Figure 3) and additionally contains two path coefficients from FC to BI and from SI to USE. Both relationships can be theoretically justified. First, in their initial presentation of UTAUT, Venkatesh et al. (2003) assume that FC serves as a proxy for actual behavioral control and, therefore, influences behavior directly. However, in their extension of the original model, Venkatesh et al. (2012)) note that FC can vary considerably across contexts. Hence, FC "will act more like perceived behavioral control in the theory of planned behavior (TPB) and influence both intention and behavior." (p. 162). Our results provide support for this notion. Second, Blut et al.'s (2022) recent meta-analysis of the UTAUT model drawing on 1,935 samples shows that SI correlates strongly with habit, which several follow-up papers to Venkatesh et al. (2003) have considered an additional antecedent to USE. In the original model used in our analysis (i.e., without habit), SI may absorb some of habit's direct impact on USE, which was very pronounced in Blut et al.'s (2022) analysis. Alternatively, coercive pressure may explain SI's direct effect on USE. Specifically, a number of sources (e.g., regulatory agencies) may generate formal or informal coercive pressure as a type of social influence, thereby triggering use behavior (Bozan et al., 2016). Tables 2 and 3 provide the component weight, loading, and path coefficient estimates of the selected model. All the parameter estimates are statistically significant.

Table 4 shows how the PF algorithm is carried out for the example. In Step 1, the algorithm begins by treating the null model (Model 0), which includes no predictors for the target components, as M_1 and calculates its $CVPE_T$ value. In Step 2, the algorithm considers nine competing models (Models 1 to 9), each of which adds one path coefficient to M_1. As Model 9 shows the smallest $CVPE_T$, it is considered M_2 against M_1. The PF algorithm then conducts a statistical test for the $CVPE_T$ difference between M_1 and M_2. As the $CVPE_T$ value of M_2 is statistically significantly smaller than that of M_1, Model 9 is chosen over Model 0 and becomes M_1. The algorithm repeats Step 2 for the remaining candidate path coefficients until the addition of a path coefficient

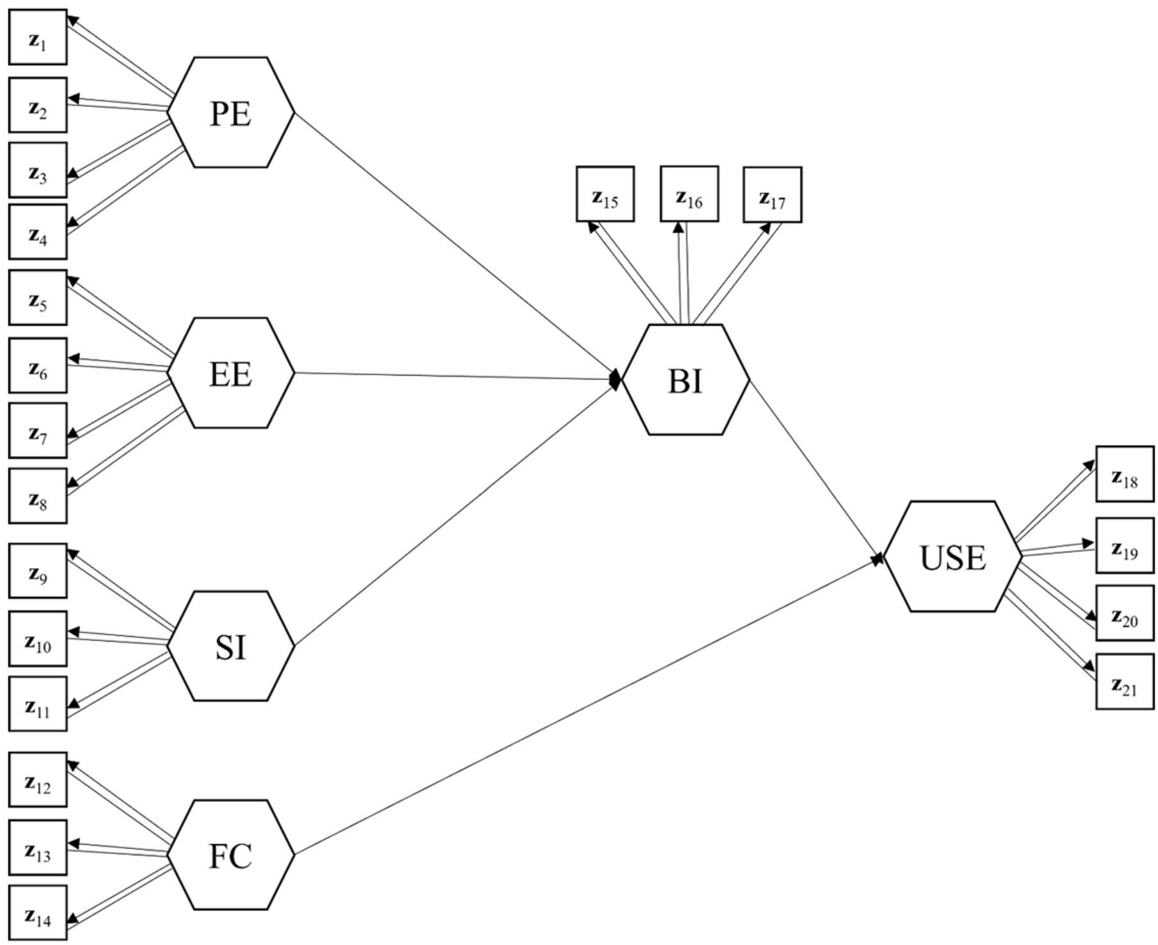

Figure 3. The unified theory of acceptance and use of technology (UTAUT) model. *Note:* Hexagons and squares denote components and indicators, respectively. Straight lines indicate the weights and arrows signify the loadings or path coefficients in the original UTAUT model. Error terms are omitted.

does not lead to a statistically significant difference in the $\mathrm{CVPE_T}$ values. In this way, the algorithm chooses Model 40 as the final model with the highest prediction accuracy for the target components.

5. Concluding Remarks

We proposed a data-driven specification search algorithm for GSCA, named the predictive feedforward search algorithm, which can be used when little theoretical knowledge is available for fully specifying the relationships among variables. Notably, this algorithm is prediction-oriented in that it aims to find a model with the highest prediction power, without recourse to distributional assumptions. Our simulation study showed that the final models selected via the algorithm tended to be more parsimonious ones that identify the true predictors mostly when sample size was small, while they converged to the pre-specified model when sample size became large. Conversely, the tabu search algorithm failed to filter out false predictors across sample sizes, though it showed a higher power for detecting the true predictors than the proposed algorithm when sample size was small or moderate. In our real data analysis, both search algorithms favored a more complex version of the UTAUT model as the best predictive model, which additionally included the effect of SI on USE and the effect of FC on BI. The addition of these additional effects to the UTAUT model has been empirically discussed in other studies (Bozan et al., 2016; Venkatesh et al., 2003).

Model specification searches can be extremely complex, especially when the number of possible structural relationships is large. The proposed algorithm can serve as an efficient tool for prediction-oriented specification search in GSCA. The approach conducts a directed specification search in that it uses a sequence of predictors to identify the candidate model with the lowest prediction error—it does not consider all potential structural relationships. Researchers have proposed heuristics such as ant optimization (Marcoulides & Drezner, 2003) and genetic algorithm (Marcoulides & Drezner, 2001), which future research may generalize to GSCA. Nevertheless, a local optimization procedure as proposed in this research proves particularly useful for improving models that are not fundamentally misspecified but are incorrect only to the extent that they have some missing paths or are overspecified (e.g., as in our empirical UTAUT model example). Future research may also extend our simulation study by considering more layers of variables. Doing so would allow for testing the algorithm's efficacy for detecting (serial) mediating relationships. Additional studies could extend the model by considering nonlinear or other types of interaction effects.

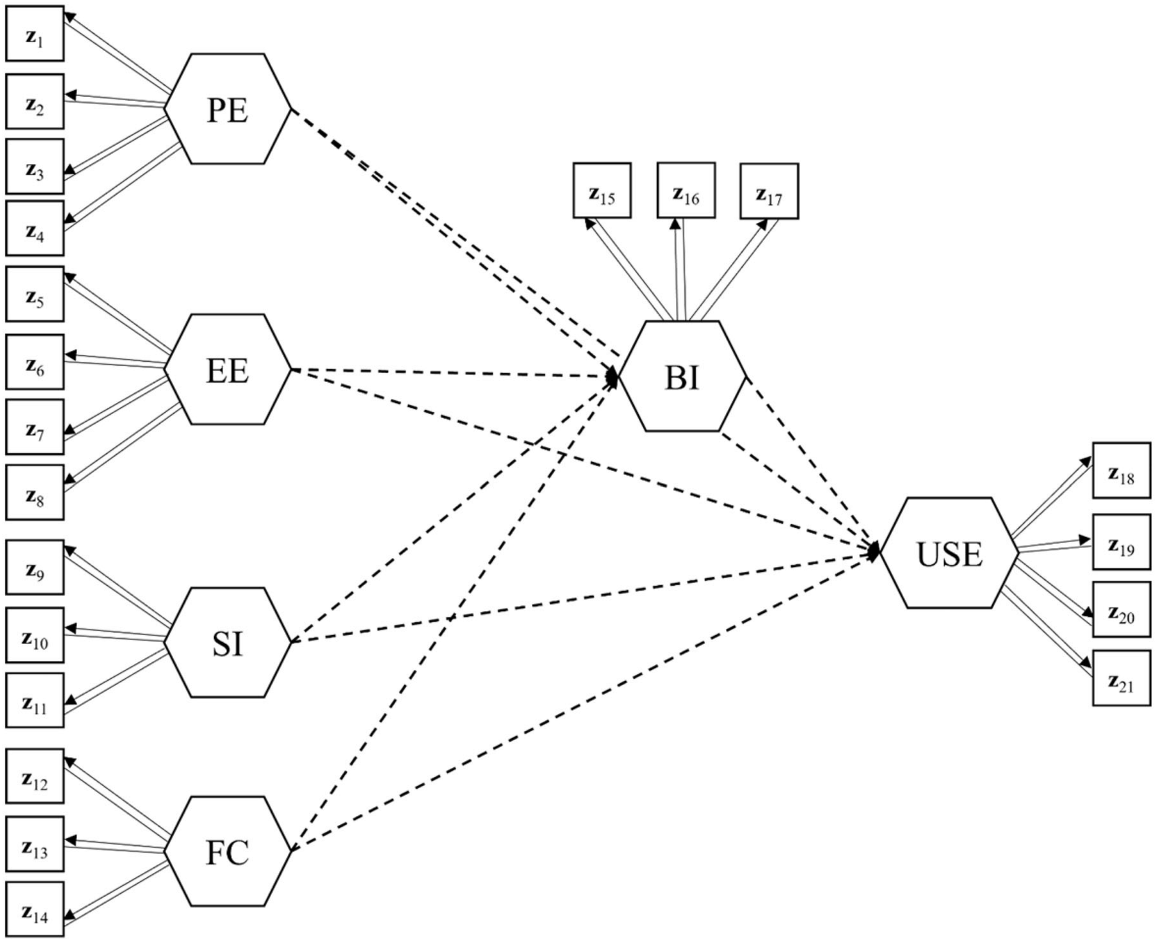

Figure 4. The relationships among the UTAUT components considered by the proposed and tabu search algorithms. *Note:* Hexagons and squares denote components and indicators, respectively. Straight lines indicate the weights and arrows signify the loadings or path coefficients in the default model that the algorithm starts with, whereas dashed arrows represent path coefficients that the algorithm will consider for selection. Error terms are omitted.

Table 2. The loading and weight estimates and their standard errors (SE) and 95% confidence intervals (CI) in the final UTAUT model selected via the proposed and tabu search algorithms.

		Weights			Loadings		
Component	Indicator	Estimate	SE	95% CI	Estimate	SE	95% CI
PE	z_1	.282	.008	[.264, .297]	.803	.028	[.739, .849]
	z_2	.321	.010	[.303, .343]	.888	.013	[.859, .911]
	z_3	.317	.010	[.300, .340]	.890	.012	[.864, .911]
	z_4	.273	.007	[.259, .288]	.755	.028	[.693, .804]
EE	z_5	.306	.008	[.291, .322]	.837	.015	[.805, .864]
	z_6	.297	.008	[.283, .313]	.805	.018	[.768, .840]
	z_7	.295	.007	[.282, .309]	.842	.019	[.799, .877]
	z_8	.297	.009	[.280, .314]	.862	.016	[.829, .889]
SI	z_9	.354	.009	[.337, .372]	.938	.008	[.922, .951]
	z_{10}	.351	.010	[.330, .370]	.948	.007	[.935, .961]
	z_{11}	.367	.007	[.353, .380]	.912	.012	[.885, .933]
FC	z_{12}	.328	.029	[.254, .373]	.507	.054	[.378, .597]
	z_{13}	.498	.016	[.471, .530]	.848	.015	[.818, .879]
	z_{14}	.493	.017	[.461, .525]	.834	.016	[.801, .866]
BI	z_{15}	.432	.026	[.384, .483]	.739	.028	[.680, .788]
	z_{16}	.468	.030	[.408, .526]	.696	.038	[.613, .758]
	z_{17}	.492	.026	[.442, .544]	.723	.035	[.645, .784]
USE	z_{18}	.351	.013	[.326, .375]	.804	.016	[.771, .833]
	z_{19}	.317	.014	[.291, .343]	.769	.018	[.732, .800]
	z_{20}	.331	.011	[.311, .355]	.777	.016	[.745, .804]
	z_{21}	.307	.012	[.283, .328]	.705	.020	[.664, .741]

Table 3. The path coefficient estimates and their standard errors (SE) and 95% confidence intervals (CI) in the final UTAUT model selected via the proposed and tabu search algorithms.

Path	Estimate	SE	95% CI
PE → BI	.191	.042	[.111, .281]
EE → BI	.302	.039	[.224, .378]
SI → BI	.189	.038	[.111, .258]
FC → BI	.195	.035	[.124, .262]
SI → USE	−.169	.042	[−.253, −.086]
FC → USE	.185	.038	[.105, .255]
BI → USE	.453	.042	[.368, .532]

While the proposed algorithm offers a means to identify an appropriate model specification from a prediction standpoint, additional steps need to be taken in order to safeguard the model's validity from explanation perspectives. Above all, researchers need to provide strong theoretical justification about the model selected by the proposed algorithm, if they are interested in a causal interpretation of the model. The proposed algorithm focuses solely on searching for the best combination of predictors that minimizes prediction errors for target variables. This does not guarantee that the chosen predictors are also causal for the target variables. If researchers mistakenly omit a variable that actually causes target variables from a list of candidate predictors, the algorithm may select a false predictor that is simply associated with the omitted variable, tending to overestimate its effect (Angrist & Pischke, 2009, Chapter 3). Moreover,

Table 4. Candidate UTAUT models considered in the proposed search algorithm.

# of paths included	Model number	$CVPE_T$	Additional path considered			M_1
0	0	1				v
1	1	0.905	PE	to	BI	
1	2	0.884	EE	to	BI	
1	3	0.934	SI	to	BI	
1	4	0.921	FC	to	BI	
1	5	0.982	PE	to	USE	
1	6	0.960	EE	to	USE	
1	7	1.002	SI	to	USE	
1	8	0.933	FC	to	USE	
1	9	0.865	BI	to	USE	v
2	10	0.792	PE	to	BI	
2	11	0.764	EE	to	BI	v
2	12	0.821	SI	to	BI	
2	13	0.792	FC	to	BI	
2	14	0.865	PE	to	USE	
2	15	0.867	EE	to	USE	
2	16	0.858	SI	to	USE	
2	17	0.852	FC	to	USE	
3	18	0.740	PE	to	BI	v
3	19	0.741	SI	to	BI	
3	20	0.741	FC	to	BI	
3	21	0.764	PE	to	USE	
3	22	0.765	EE	to	USE	
3	23	0.756	SI	to	USE	
3	24	0.750	FC	to	USE	
4	25	0.726	SI	to	BI	
4	26	0.723	FC	to	BI	v
4	27	0.740	PE	to	USE	
4	28	0.741	EE	to	USE	
4	29	0.732	SI	to	USE	
4	30	0.725	FC	to	USE	
5	31	0.712	SI	to	BI	
5	32	0.723	PE	to	USE	
5	33	0.724	EE	to	USE	
5	34	0.714	SI	to	USE	
5	35	0.707	FC	to	USE	v
6	36	0.696	SI	to	BI	
6	37	0.706	PE	to	USE	
6	38	0.710	EE	to	USE	
6	39	0.696	SI	to	USE	v
7	40	0.685	SI	to	BI	v
7	41	0.697	PE	to	USE	
7	42	0.698	EE	to	USE	
8	43	0.686	PE	to	USE	
8	44	0.687	EE	to	USE	

Note: $CVPE_T$ = the cross-validation estimate of expected prediction error for target variables; the last column indicates which candidate models have ever been considered the baseline model M_1 during the execution of the algorithm; the last M_1 (Model 40) is the final model selected by the algorithm.

even when all causal predictors for target variables are considered, the algorithm may select only a subset of them because a simpler model can have higher prediction power than the true model in a finite sample condition (Shmueli, 2010). Cross-validating and replicating the model selected by the algorithm in different contexts and with different samples would be mandatory to support its causal interpretability and generalizability.

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